

DIEGO ZIDANE REYES GOMEZ

San Diego, California

☎ (818)-671-9714

✉ dreyesgomez@ucsd.edu

🐙 github.com/dreyes32

🌐 linkedin.com/in/diegozrg-dev

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, HTML/CSS, R, Kotlin

AI/ML: PyTorch, RAG, LLM Agents, Prompt Engineering, Vector Search, Model Evaluation

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Playwright

Cloud/DevOps: AWS, S3, Docker, Linux, Git, REST APIs, MCP, Postman

Developer Tools: VS Code, Cursor, Unreal Engine 5, Android Studio, Figma, Adobe Creative Cloud

Professional Experience

Werfen

June 2026 – September 2026

AI Engineer Intern (Marketing & Digital Innovation Team)

San Diego, CA

- Developed and productionized a RAG-powered Marketing Agent within WALT, Werfen's enterprise multi-agent AI platform, automating technical affiliate support for an initial portfolio of 5 diagnostic product lines while building the connector and retrieval architecture to scale across additional Strategic Diagnostics, Blood Gas, and worldwide product portfolios.
- Engineered document-grounded retrieval, intent classification, iterative clarification, escalation, and email-generation workflows using validated SAP documentation; implemented connector logic that preserves regional and language-specific document variants, enabling WALT to retrieve market-appropriate technical information while maintaining traceable, supported responses.
- Developed an IT support agent for WALT with knowledge-grounded troubleshooting, clarification logic, duplicate-query prevention, escalation handling, and ServiceNow integration for securely retrieving incident and request status based on authenticated user access.
- Built MCP tooling for WALT agents, including AWS S3-based part-number image retrieval with structured tool outputs and image-availability detection, as well as ServiceNow MCP functionality for authenticated incident and requested-item status retrieval.
- Improved enterprise knowledge ingestion by implementing SAP document-type exclusions, case-insensitive connector handling, document-family matching, and resilient multi-connector synchronization with safeguards against partial-sync data deletion and failed source retrievals.
- Diagnosed and resolved production connector failures by implementing refresh-and-retry logic for expired SAP download URLs and investigating duplicate document processing across AWS-backed knowledge libraries.
- Expanded WALT platform reliability through Python/TypeScript development, FastAPI APIs, Postman validation, regression testing, and Playwright E2E debugging while collaborating with Marketing and Digital Innovation stakeholders to translate requirements into validated production workflows.

San Diego Supercomputer Center (SDSC)

June 2023 – June 2026

Software Engineer Intern

September 2023 - June 2026

- Redesigned *Workload*, a full-stack work management platform for SDSC's Research Data Services department, using Next.js, React, TypeScript, and Tailwind CSS to centralize team assignments and project activity across GitLab, Zendesk, and Trello.
- Built a responsive Kanban workflow with member selection, project-specific views, issue filtering, and status-based task organization, improving visibility into active, blocked, pending, completed, and uncategorized work across multiple external platforms.
- Containerized the Next.js frontend, Go backend, and PostgreSQL database with Docker Compose and configured API endpoints and environment variables to support reliable local development and dynamic data retrieval across integrated services.

Frontend Android Developer Intern

June 2023 - September 2023

- Developed *Bloom Book*, an Android application for tracking plant growth and development, building responsive mobile interfaces and data-driven visualizations for recording and reviewing plant progress over time.
- Implemented interactive application features in Kotlin using Android Studio and integrated persistent data storage to support structured plant-growth records and user-generated tracking data.
- Designed wireframes and interactive prototypes in Figma to define user flows, information architecture, and interface behavior before translating designs into the production Android application.

Academic & Research Experience

Qualcomm Institute – UCSD Serious Games Lab

September 2025 – June 2026

Software Engineer Intern (Game Systems & VR Development) – The Climate Games

San Diego, CA

- Built gameplay logic and visualization pipelines using Unreal Engine and C++ for real time interactive systems.
- Implemented a modular fire system to model ignition, propagation, extinguishing mechanics, and damage over time.
- Improved runtime performance by refactoring Blueprint-heavy gameplay systems into reusable components backed by native C++ logic.
- Developed and integrated Niagara based particle systems for fluid fire and smoke VFX, leveraging GPU particle simulation and parameter driven emitters for scalable environmental effects.

UCSD Cognitive Science Department

June 2024 – January 2026

UE Developer & Research Assistant, Systems Neuroscience Lab

January 2025 - January 2026

- Supported Unreal Engine development for VR experiments investigating self-motion and trajectory perception.
- Architected immersive experimental environments with controlled motion parameters, stimulus timing systems, and data-logging pipelines for behavioral analysis.

Instructional Assistant & Programming Tutor

June 2024 - January 2025

- **COGS 3 (Introduction to Computing):** Guided students through foundational web development concepts and assignments using HTML, CSS, and JavaScript.
- **COGS 18 (Introduction to Python):** Led weekly programming labs, supported students with Python assignments, proctored coding-based exams, and developed a Google Apps Script to identify students needing additional support.
- **COGS 187A (Usability & Information Architecture):** Mentored students in usability, information architecture, and interface design using Figma, including interactive systems for XR environments.
- **COGS 160 (Metabolic Health Analytics):** Assisted students with data processing, visualization, and time-series analysis in R for metabolic health applications.

UCSD Engineering Department - Global TIES

September 2024 – March 2025

UX Researcher, Cognovate Labs (Med-Tech Startup)

San Diego, CA

- Collaborated with EMTs and healthcare professionals to identify gaps in pre-hospital stroke assessment, addressing delays in critical medical decision making.
- Designed an EEG-based neural signal processing system with UX driven workflows to provide real time neurological insights for first responders.
- Researched electrocardiography (EKG) integration to enhance diagnostic accuracy and reduce latency in emergency stroke evaluation.

UCSD Computer Science & Engineering Department - ERSP

September 2023 – June 2024

Undergraduate ML Researcher - McAuley Lab

San Diego, CA

- Developed an LSTM-based sequence model for score-to-performance rendering, predicting expressive onset and offset timing from symbolic violin sheet music using the Bach aligned MIDI-performance dataset.
- Built preprocessing and PyTorch training pipelines to encode sheet music as structured onset, duration, and pitch features mapped to ground-truth expressive timing targets.
- Implemented a custom Euclidean-distance loss and benchmarked linear regression, MLP, and LSTM architectures for modeling temporal dependencies in expressive violin performance.
- Generated playable MIDI from predicted expressive timing using MusPy for qualitative comparison against human performances.

Honors & Research Programs

UC San Diego Early Research Scholars Program (ERSP)

June 2024

UCSD Computer Science & Engineering (CSE) Department

University of California San Diego

- Completed UCSD's competitive three quarter ERSP, receiving graduate level research training under CSE faculty.
- Collaborated with a small team to design, build, and present an original computer science research project.

UC San Diego McNair Scholars Program

November 2025 – August 2026

McNair Scholar - 3D Machine Learning Research

University of California San Diego

- Designed a learning-based framework for adaptive 3D mesh optimization in XR systems, representing meshes as graph structures and applying Graph Neural Networks (GNNs) to predict vertex level importance for dynamic refinement and simplification.
- Developed a real time pipeline integrating geometric feature extraction (curvature, normals, visibility) with PyTorch Geometric and Unreal Engine to balance visual fidelity and computational efficiency under runtime constraints.

Education

University of California San Diego

September 2022 – June 2026

B.S. Cognitive Science (Human-Computer Interaction), Minor in Computer Science

La Jolla, San Diego, CA